4	**(a)**	$$\frac{R}{0.75+R+5.5}\times 4.0 = 1.3$$ $$R = 3.0\ \Omega$$	C1
		$$\text{p.d. across AB} = \frac{5.5}{0.75+3.0+5.5}\times 4.0$$	C1
		$$= 2.4\ \text{V}$$	A0
	(b)(i)	$$E = \frac{1.50-0.56}{1.50}\times 2.4$$	C1
		$$= 1.5\ \text{V}$$	A1
	(b)(ii)	As C is shifted closer to A, the potential at C increases, thus increasing the potential difference between BC.	M1
		Since the potential between BC will become larger than the terminal p.d. of cell Q, current will now flow from C to B through cell Q.	A1
	(c)(i)	$$\frac{R}{L} = 5.5/150 = 0.0367\ \Omega\ \text{cm}^{-1}$$	C1
		$$R = 2\times 15\times 0.0367 + \frac{1}{5}\left(\frac{150-2\times 15}{5}\times 0.0367\right)$$	C1
		$$= 1.28\ \Omega$$ $$= 1.3\ \Omega$$	A0
	(c)(ii)	Since $I = n A v q$, number density n, cross-sectional area A and charge q are the same in both sections XY (consider a single wire) and AX, $v \propto I$. the current through a single wire in XY is 1/5 of the current through AX. OR Since the same current flows through sections AX and XY (consider XY as a whole), $v \propto 1/A$. the cross-sectional area of XY is 5 times the cross-sectional area of AX.	C1
		drift velocity v is greater in AX than in XY	A1